SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - Database Design and Connectivity

Course Code:	DSA0501	**Course Title:**	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO2	**Assessment:**	**AT1 – Database Design and Connectivity**
Weightage:		**Total Marks:**	20 Marks
CO2 Statement: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)			
Student Name:	**U.Ananya**	**Reg. No.:**	**192424382**
Section / Batch:	**AI&DS/24 Batch**	**Date of Submission:**	**29-07-2026**
Faculty In charge:	K Veena devi	**Project Mode:**	Individual Submission

Code of Conduct

I U.Ananya(192424382) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: __U.Ananya__

Requirement Analysis (4 Marks)	ER Diagram Design (4 Marks)	Table Design & Normalization (4 Marks)	Database Connectivity using Python (4 Marks)	CRUD Operations (4 Marks)	Total (20 Marks)

CO2 – AT1 – Database Design and Connectivity

Sample Questions

1. Student Database Design and Connectivity

Python Code:

```
import sqlite3

# Connect to SQLite database

conn = sqlite3.connect("student.db")

# Create cursor object

cur = conn.cursor()

# Create Student table

cur.execute("""

CREATE TABLE IF NOT EXISTS Student(

    RegNo INTEGER PRIMARY KEY,

    Name TEXT,

    Department TEXT,

    Age INTEGER,

    CGPA REAL

)

""")

# Insert 5 student records

students = [

    (101, "Ananya", "AI&DS", 19, 9.2),

    (102, "Rahul", "CSE", 20, 8.8),

    (103, "Priya", "ECE", 19, 9.1),

    (104, "Kiran", "IT", 20, 8.5),

    (105, "Sneha", "AI&DS", 19, 9.4)
```

```
]

cur.executemany("INSERT OR IGNORE INTO Student VALUES(?,?,?,?,?)", students)

# Save changes

conn.commit()

# Display all records

print("Student Records")

print("-" * 50)

cur.execute("SELECT * FROM Student")

rows = cur.fetchall()

for row in rows:

    print(row)

# Close connection

conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/student databa
se.py
Student Records
--------------------------------------------------
(101, 'Ananya', 'AI&DS', 19, 9.2)
(102, 'Rahul', 'CSE', 20, 8.8)
(103, 'Priya', 'ECE', 19, 9.1)
(104, 'Kiran', 'IT', 20, 8.5)
(105, 'Sneha', 'AI&DS', 19, 9.4)
>>> |
```

2. Library Management Database

Python Code:

```
import sqlite3

conn = sqlite3.connect("library.db")
```

```
cur = conn.cursor()

cur.execute("""

CREATE TABLE IF NOT EXISTS Book(

    BookID INTEGER PRIMARY KEY,

    Title TEXT,

    Author TEXT,

    Publisher TEXT,

    Price REAL

books = [

    (101, "Python Basics", "John", "ABC Publications", 450),

    (102, "Data Science", "David", "XYZ Publications", 550),

    (103, "Machine Learning", "James", "Tech Books", 700),

    (104, "DBMS", "Priya", "Pearson", 500),

    (105, "Operating Systems", "Kumar", "McGraw Hill", 650)

]

cur.executemany("INSERT OR IGNORE INTO Book VALUES(?,?,?,?,?)", books)

cur.execute("UPDATE Book SET Price = 750 WHERE BookID = 103")

conn.commit()
```

```
cur.execute("SELECT * FROM Book WHERE BookID = 103")

print("Updated Book Details")

print("-" * 50)

for row in cur.fetchall():

    print(row)

conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/book details.p
y
Updated Book Details
--------------------------------------------------
(103, 'Machine Learning', 'James', 'Tech Books', 750.0)
```

3. Employee and Department Database Design

Python Code:

```
import sqlite3

# Connect to database

conn = sqlite3.connect("company.db")

cur = conn.cursor()

# Create Department table

cur.execute("""

CREATE TABLE IF NOT EXISTS Department(

    DeptID INTEGER PRIMARY KEY,

    DeptName TEXT
```

```
)
""")
# Create Employee table
cur.execute("""
CREATE TABLE IF NOT EXISTS Employee(
    EmpID INTEGER PRIMARY KEY,
    EmpName TEXT,
    Salary REAL,
    DeptID INTEGER,
    FOREIGN KEY(DeptID) REFERENCES Department(DeptID)
)
""")
# Insert Department records
departments = [
    (1, "HR"),
    (2, "Finance"),
    (3, "IT"),
    (4, "Marketing")
]
cur.executemany("INSERT OR IGNORE INTO Department VALUES(?,?)", departments)
# Insert Employee records
employees = [
    (101, "Ananya", 50000, 3),
    (102, "Rahul", 45000, 2),
    (103, "Priya", 55000, 1),
    (104, "Kiran", 48000, 4),
    (105, "Sneha", 60000, 3)
```

```
]
cur.executemany("INSERT OR IGNORE INTO Employee VALUES(?,?,?,?)", employees)
# Save changes
conn.commit()
# JOIN query
print("Employee Name\tDepartment")
cur.execute("""
SELECT Employee.EmpName, Department.DeptName
FROM Employee
INNER JOIN Department
ON Employee.DeptID = Department.DeptID
""")
rows = cur.fetchall()
for row in rows:
    print(row[0], "\t", row[1])
conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/department nam
es.py
Employee Name   Department
Ananya   IT
Rahul    Finance
Priya    HR
Kiran    Marketing
Sneha    IT
```

4. Hospital Patient Database

Python Code:

```
import sqlite3
# Connect to SQLite database
```

```
conn = sqlite3.connect("hospital.db")

# Create cursor

cur = conn.cursor()

# Create Patient table

cur.execute("""

CREATE TABLE IF NOT EXISTS Patient(

    PatientID INTEGER PRIMARY KEY,

    PatientName TEXT,

    Age INTEGER,

    Disease TEXT,

    ContactNo TEXT

)

""")

# Insert patient records

patients = [

    (101, "Rahul", 35, "Fever", "9876543210"),

    (102, "Priya", 28, "Diabetes", "9876543211"),

    (103, "Arjun", 45, "Asthma", "9876543212"),

    (104, "Sneha", 30, "Dengue", "9876543213"),

    (105, "Kiran", 40, "Typhoid", "9876543214")

]

cur.executemany("INSERT OR IGNORE INTO Patient VALUES(?,?,?,?,?)", patients)

# Update contact number of Patient ID 103

cur.execute("""

UPDATE Patient

SET ContactNo = '9999999999'

WHERE PatientID = 103
```

```
""")
# Delete discharged patient (Patient ID 104)
cur.execute("""
DELETE FROM Patient
WHERE PatientID = 104
""")
# Save changes
conn.commit()
# Display remaining records
print("Remaining Patient Records")
print("-" * 60)
cur.execute("SELECT * FROM Patient")
rows = cur.fetchall()
for row in rows:
    print(row)
# Close connection
conn.close()
```

Output:

```
>>>
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/patient record
s.py
Remaining Patient Records
------------------------------------------------------------
(101, 'Rahul', 35, 'Fever', '9876543210')
(102, 'Priya', 28, 'Diabetes', '9876543211')
(103, 'Arjun', 45, 'Asthma', '9999999999')
(105, 'Kiran', 40, 'Typhoid', '9876543214')
>>>
```

5. Online Course Registration System

Python Code:

```
import sqlite3
```

```
# Connect to SQLite database

conn = sqlite3.connect("course_registration.db")

# Create cursor

cur = conn.cursor()

# Create Student table

cur.execute("""

CREATE TABLE IF NOT EXISTS Student(

    StudentID INTEGER PRIMARY KEY,

    StudentName TEXT,

    Department TEXT

)

""")

# Create Course_Registration table

cur.execute("""

CREATE TABLE IF NOT EXISTS Course_Registration(

    CourseID INTEGER PRIMARY KEY,

    CourseName TEXT,

    StudentID INTEGER,

    FOREIGN KEY(StudentID) REFERENCES Student(StudentID)

)

""")

# Insert Student records

students = [

    (101, "Ananya", "AI&DS"),

    (102, "Rahul", "CSE"),

    (103, "Priya", "ECE"),

    (104, "Kiran", "IT"),
```

```
    (105, "Sneha", "AI&DS")
]
cur.executemany("INSERT OR IGNORE INTO Student VALUES(?,?,?)", students)
# Insert Course Registration records
courses = [
    (1, "Python Programming", 101),
    (2, "Data Science", 102),
    (3, "Machine Learning", 103),
    (4, "Database Systems", 104),
    (5, "Artificial Intelligence", 105)
]
cur.executemany("INSERT OR IGNORE INTO Course_Registration VALUES(?,?,?)", courses)
# Save changes
conn.commit()
# Display Student Name and Course Name
print("Student Name\t\tCourse Name")
cur.execute("""
SELECT Student.StudentName, Course_Registration.CourseName
FROM Student
INNER JOIN Course_Registration
ON Student.StudentID = Course_Registration.StudentID
""")
rows = cur.fetchall()
for row in rows:
    print(row[0], "\t\t", row[1])
# Close connection
conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/course registration.py
Student Name            Course Name
Ananya           Python Programming
Rahul            Data Science
Priya            Machine Learning
Kiran            Database Systems
Sneha            Artificial Intelligence
```

6. Banking Account Management System

Python Code:

```
import sqlite3
# Connect to SQLite database
conn = sqlite3.connect("bank.db")
# Create cursor
cur = conn.cursor()
# Create Account table
cur.execute("""
CREATE TABLE IF NOT EXISTS Account(
    AccountNo INTEGER PRIMARY KEY,
    CustomerName TEXT,
    AccountType TEXT,
    Balance REAL,
    Branch TEXT
)
""")
# Insert customer records
accounts = [
    (1001, "Rahul", "Savings", 25000, "Chennai"),
    (1002, "Priya", "Current", 50000, "Hyderabad"),
    (1003, "Ananya", "Savings", 30000, "Bangalore"),
    (1004, "Kiran", "Current", 45000, "Mumbai"),
    (1005, "Sneha", "Savings", 28000, "Delhi")
]
cur.executemany("INSERT OR IGNORE INTO Account VALUES(?,?,?,?,?)", accounts)
# Update balance of Account No. 1003
cur.execute("""
UPDATE Account
SET Balance = 35000
WHERE AccountNo = 1003
""")
# Save changes
```

```
conn.commit()
# Display all records
print("Customer Account Details")
print("-" * 70)
cur.execute("SELECT * FROM Account")
rows = cur.fetchall()
for row in rows:
    print(row)
# Close connection
conn.close()
```

Output:

```
==================================================== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/bank details.py
Customer Account Details
----------------------------------------------------------------------
(1001, 'Rahul', 'Savings', 25000.0, 'Chennai')
(1002, 'Priya', 'Current', 50000.0, 'Hyderabad')
(1003, 'Ananya', 'Savings', 35000.0, 'Bangalore')
(1004, 'Kiran', 'Current', 45000.0, 'Mumbai')
(1005, 'Sneha', 'Savings', 28000.0, 'Delhi')
```

7. Inventory Management System

Python Code:

```
import sqlite3

conn = sqlite3.connect("inventory.db")

cur = conn.cursor()

cur.execute("""

CREATE TABLE IF NOT EXISTS Product(

    ProductID INTEGER PRIMARY KEY,

    ProductName TEXT,

    Category TEXT,

    Quantity INTEGER,
```

```
    UnitPrice REAL
)
""")
products = [
    (101, "Laptop", "Electronics", 15, 55000),
    (102, "Mouse", "Accessories", 8, 500),
    (103, "Keyboard", "Accessories", 5, 1200),
    (104, "Monitor", "Electronics", 20, 12000),
    (105, "Printer", "Electronics", 7, 8500)
]
cur.executemany("INSERT OR IGNORE INTO Product VALUES(?,?,?,?,?)", products)
cur.execute("""
UPDATE Product
SET Quantity = 20
WHERE Quantity < 10
""")
conn.commit()
```

```
print("Updated Inventory")

print("-" * 60)

cur.execute("SELECT * FROM Product")

rows = cur.fetchall()

for row in rows:

    print(row)

conn.close()
```

Output:

```
>>>
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/inventory.py =
Updated Inventory
------------------------------------------------------------
(101, 'Laptop', 'Electronics', 15, 55000.0)
(102, 'Mouse', 'Accessories', 20, 500.0)
(103, 'Keyboard', 'Accessories', 20, 1200.0)
(104, 'Monitor', 'Electronics', 20, 12000.0)
(105, 'Printer', 'Electronics', 20, 8500.0)
>>>
```

8. Course and Faculty Management System

Python Code:

```
import sqlite3

# Connect to SQLite database

conn = sqlite3.connect("university.db")

# Create cursor

cur = conn.cursor()

# Create Faculty table
```

```
cur.execute("""
CREATE TABLE IF NOT EXISTS Faculty(
   FacultyID INTEGER PRIMARY KEY,
   FacultyName TEXT,
   Department TEXT
)
""")
# Create Course table
cur.execute("""
CREATE TABLE IF NOT EXISTS Course(
   CourseID INTEGER PRIMARY KEY,
   CourseName TEXT,
   FacultyID INTEGER,
   FOREIGN KEY(FacultyID) REFERENCES Faculty(FacultyID)
)
""")
# Insert Faculty records
faculty = [
   (101, "Dr. Ravi", "CSE"),
   (102, "Dr. Priya", "AI&DS"),
   (103, "Dr. Kumar", "ECE"),
   (104, "Dr. Meena", "IT"),
   (105, "Dr. Arjun", "EEE")
]
cur.executemany("INSERT OR IGNORE INTO Faculty VALUES(?,?,?)", faculty)
# Insert Course records
courses = [
```

```
    (1, "Python Programming", 101),
    (2, "Machine Learning", 102),
    (3, "Digital Electronics", 103),
    (4, "Database Systems", 104),
    (5, "Power Systems", 105)
]
cur.executemany("INSERT OR IGNORE INTO Course VALUES(?,?,?)", courses)
# Save changes
conn.commit()
# Display Faculty Name and Course Name
print("Faculty Name\t\tCourse Assigned")
cur.execute("""
SELECT Faculty.FacultyName, Course.CourseName
FROM Faculty
INNER JOIN Course
ON Faculty.FacultyID = Course.FacultyID
""")
rows = cur.fetchall()
for row in rows:
    print(row[0], "\t\t", row[1])
# Close connection
conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/course faculty
.py
Faculty Name            Course Assigned
Dr. Ravi                 Python Programming
Dr. Priya                Machine Learning
Dr. Kumar                Digital Electronics
Dr. Meena                Database Systems
Dr. Arjun                Power Systems
```

9. Vehicle Registration Database

Python Code:

```
import sqlite3

# Connect to SQLite database

conn = sqlite3.connect("vehicle.db")

# Create cursor

cur = conn.cursor()

# Create Vehicle table

cur.execute("""

CREATE TABLE IF NOT EXISTS Vehicle(

    RegNo TEXT PRIMARY KEY,

    OwnerName TEXT,

    VehicleType TEXT,

    Model TEXT,
```

```
    Year INTEGER
)
""")
# Insert vehicle records
vehicles = [
    ("TN01AB1234", "Rahul", "Car", "Hyundai i20", 2022),
    ("TN02CD5678", "Priya", "Bike", "Honda Activa", 2021),
    ("TN03EF9012", "Ananya", "Car", "Maruti Swift", 2023),
    ("TN04GH3456", "Kiran", "Truck", "Tata Ace", 2020),
    ("TN05IJ7890", "Sneha", "Scooter", "TVS Jupiter", 2022)
]
cur.executemany("INSERT OR IGNORE INTO Vehicle VALUES(?,?,?,?,?)", vehicles)
conn.commit()
# Search for a vehicle using Registration Number
reg = "TN03EF9012"
cur.execute("SELECT * FROM Vehicle WHERE RegNo=?", (reg,))
record = cur.fetchone()
```

```
print("Vehicle Details")

print("-" * 60)

if record:

    print(record)

else:

    print("Vehicle not found")

conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/vehical regist
ration.py
Vehicle Details
------------------------------------------------------------
('TN03EF9012', 'Ananya', 'Car', 'Maruti Swift', 2023)
```

10. Hospital Appointment Management System

Python Code:

```
import sqlite3

# Connect to SQLite database

conn = sqlite3.connect("hospital_appointment.db")

# Create cursor

cur = conn.cursor()

# Create Doctor table

cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS Doctor(
    DoctorID INTEGER PRIMARY KEY,
    DoctorName TEXT,
    Specialization TEXT
)
""")
# Create Appointment table
cur.execute("""
CREATE TABLE IF NOT EXISTS Appointment(
    AppointmentID INTEGER PRIMARY KEY,
    PatientName TEXT,
    AppointmentDate TEXT,
    DoctorID INTEGER,
    FOREIGN KEY(DoctorID) REFERENCES Doctor(DoctorID)
)
""")
# Insert Doctor records
doctors = [
    (101, "Dr. Ravi", "Cardiology"),
    (102, "Dr. Priya", "Neurology"),
    (103, "Dr. Kumar", "Orthopedics"),
    (104, "Dr. Meena", "Dermatology"),
    (105, "Dr. Arjun", "Pediatrics")
]
cur.executemany("INSERT OR IGNORE INTO Doctor VALUES(?,?,?)", doctors)
# Insert Appointment records
appointments = [
```

```
    (1, "Rahul", "2026-07-30", 101),

    (2, "Ananya", "2026-07-31", 102),

    (3, "Kiran", "2026-08-01", 103),

    (4, "Sneha", "2026-08-02", 104),

    (5, "Priya", "2026-08-03", 105)

]

cur.executemany("INSERT OR IGNORE INTO Appointment VALUES(?,?,?,?)",
appointments)

# Save changes

conn.commit()

# Display Doctor Name and Appointment Details

print("Doctor Name\t\tPatient Name\tAppointment Date")

cur.execute("""

SELECT Doctor.DoctorName,

       Appointment.PatientName,

       Appointment.AppointmentDate

FROM Doctor

INNER JOIN Appointment

ON Doctor.DoctorID = Appointment.DoctorID

""")

rows = cur.fetchall()

for row in rows:

    print(row[0], "\t", row[1], "\t", row[2])

# Close connection

conn.close()
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/hospital appoi
ntment.py
Doctor Name             Patient Name    Appointment Date
Dr. Ravi         Rahul   2026-07-30
Dr. Priya        Ananya          2026-07-31
Dr. Kumar        Kiran   2026-08-01
Dr. Meena        Sneha   2026-08-02
Dr. Arjun        Priya   2026-08-03
```